VeriLoom: Guidelines as an Executable Specification

Round 1 submission: The Plan

Response to the problem statement by Inkloom AI · Nexus Code

Item	Detail
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Summary

The problem. Organisations now publish through employees, agencies, freelancers and AI tools. Wrong claims, expired facts, off-brand terms and inaccessible graphics can reach the public before a reviewer notices. Ad platforms have started editing approved creative after sign-off, and most brand review still lives in PDFs, email threads and chat.

Our thesis. A brand guideline is a specification that is rarely tested, versioned or checked for ambiguity. Generic LLM reviewers also give different verdicts on the same input. VeriLoom compiles guidelines into structured, citable rules (a Brand-AST), runs deterministic checks first, grounds every factual claim in a time-bounded Fact Vault, and measures where the guidelines themselves cause disagreement. Every finding cites a source ID, the system abstains when evidence is weak, and people make every publish decision.

Three named improvements (our commitments for Rounds 2 and 3):

- **Hedge-sensitive, time-aware claim verifier.** Copy is split into atomic claims. Each claim is checked against approved facts that are still valid on today's date, and against the strength of the wording (for example "up to 30%" becoming "always 30%").
- **Intent-locked surgical rewriting.** Suggestions change as few words as possible, are scored with an Intent Preservation Metric (IPM), and appear as accept-or-reject tracked changes. Nothing is applied automatically.
- **Measured visual and accessibility checks.** Colour distance (CIEDE2000) and WCAG 2.1 contrast are computed with exact formulas, including contrast measured under the text itself, instead of asking a vision model to guess.

Scope. The MVP covers rich text and Markdown, PNG/JPG images and PDF documents inside one web workbench, with a four-role review workflow. Live ad-platform monitoring, video, audio and external integrations are roadmap items (section 7.6).

Evidence standard. Every figure is either traceable to a named source (Appendix B), produced by our own test (section 3.2), or labelled as a hypothesis. Section 3.1 says exactly what research was and was not done, so the reader can weigh it.

1. Problem definition and evidence

1.1 The problem

We split the problem into three kinds, because each needs a different fix:

- **Content errors:** a specific piece is wrong, off-brand or inaccessible.
- **Process failures:** review exists but misses the error, or the content changes after review.

- **Specification failures:** the guidelines are unclear, outdated or contradictory, so reviewers cannot agree on what correct means.

Most brand tools address the first kind. We think much of the damage sits in the second and third kinds. That is a hypothesis (H5 in section 7.4) that we test in Round 2, not a finding.

1.2 Taxonomy of brand failures

Failure type	Typical example	How we detect it	Detectable from content alone?
Factual error	Wrong price, date, specification or attribution	Retrieval against approved facts	Yes, if an approved fact exists
Stale fact	Was true, now expired	Fact validity window (valid_from / valid_until)	Yes
Unsupported claim	"50% faster", "best", health or finance claims	Atomic claim extraction and evidence lookup	Yes
Modality inflation	"Up to 30%" published as "30%"	Compare claim strength with the approved fact	Yes
Terminology / tone drift	Banned or non-preferred terms; off-voice copy	Term matching plus semantic review	Partly
Visual misuse	Off-palette colour, distorted logo	Deterministic image checks (CIEDE2000, aspect ratio)	Yes
Accessibility gap	Missing alt text; low contrast	WCAG 2.1 maths	Yes
Post-approval drift	A platform alters content after sign-off	Compare with the live version (future scope)	No; needs the live version
Unclear guideline	Reviewers disagree because a rule is vague	Rule linting and override analytics	No; a flaw in the specification

1.3 Documented incidents

Each incident was checked against the most complete source we could access. The status column says how far that check went. Where a claim rests on one party's own account, we say so.

Incident	What the sources report	Failure type	Status
Coca-Cola "Classic" campaign (Apr-May 2025) [R1]	The ad appears to credit a non-existent book to J. G. Ballard. The agency (VML) said AI was used only to find books and that it fact-checked manually. That account is the agency's own.	Factual error that passed review (as the agency describes it)	Fortune article read in full. Agency claim is attributed, not verified.
Equinox "Question Everything But Yourself" (Jan-Jul 2026) [R3]	A campaign image drew months of criticism for stereotyping East Asian women. Commenters asked how it cleared internal review. Removed in late July 2026.	Tone and inclusion; review failure	AthleTech News (29 Jul 2026) read. The review question is attributed to commenters.
Meta AI features and advertiser liability (Oct-Nov 2026) [R2]	Meta's revised commercial terms add an AI Features section. Reports say advertisers are solely responsible for AI features that create or change ads. Bookshop.org said an account update enrolled it in a feature that added an illustration to a human-made ad. Reported effective date moved to 20 November 2026.	Post-approval drift; liability	Several trade reports read. Meta did not directly address features advertisers never switched on.
Callaway and Good Good (Aug 2026) [R13]	A creator video was produced by one party and signed off by another. Callaway's CEO said the approval should never have happened, and the partnership ended.	Approval failure; no risk routing	Adweek column read in part (paywalled).

1.4 Scale evidence

Statistic	Source	Caveat
95.9% of the top one million home pages had detectable WCAG 2 failures in 2026. Low-contrast text appeared on 83.9% and missing alt text on 53.1%.	WebAIM Million 2026 [R6]	Home pages only, not marketing assets. Confirm against the report before final export.

Statistic	Source	Caveat
In a dipstick study of Forbes India's Top 100 Digital Stars, 76% failed disclosure norms, up from 69% the year before.	ASCI Half-Yearly Complaints Report 2025-26, press release 11 Nov 2025 [R9]	A study of 100 named creators. Self-regulatory body data. ASCI guidelines are a benchmark and not binding in law for most digital advertising.
Identical LLM judge comparisons repeated 50 times flipped verdict 13.6% of the time on average; rephrased prompts changed the majority outcome in 25% of tested cases.	Yagubyan, arXiv:2606.13685 [R5]	Studies response quality, not brand compliance. Two models, one provider, single-author preprint.

1.5 What the evidence shows

- **Manual checking alone is not enough.** In the Coca-Cola case the agency describes a manual fact-check. The Equinox image cleared internal review and was removed only after months of criticism. Reviewers need findings they can trace to an approved source quickly.
- **Approval is not publication.** Platforms can change content after approval. Governance needs a record of what was approved and a way to compare it later with what ran. That comparison is future scope because it needs live platform access.
- **Unclear rules are a likely cause of inconsistent reviews.** This is our hypothesis. The evidence in section 3 is consistent with it but does not test it.

2. Target users, needs and organisational setting

2.1 Where the product is used

- In-house marketing or communications teams publishing social, email, web and decks, working with several agencies and freelancers.
- Agencies or studios serving several client brands, each with its own guidelines and approvers.
- Institutions with many non-specialist communicators publishing under one identity, such as universities.

2.2 Personas

Role	Goal	Pain today	What VeriLoom gives them
Creator (employee, freelancer or agency)	Publish quickly without being off-brand	Long PDF guidelines; late and vague rejections; rewrites that erase their voice	A pre-submit check that cites the exact rule and offers small, intent-preserving fixes
Brand or marketing manager	Consistent output across the team	Repetitive line-by-line policing of terms and colours; no view of patterns	Pre-screened queue, side-by-side diffs, recurring-issue insights
Authorised reviewer (legal, compliance)	Protect the organisation from risky claims	Must read everything; no evidence trail	Only escalated or uncertain items, with cited sources and an audit log
Brand admin	Own guidelines, approved facts and assets	Cannot tell which rules are unclear or ignored	Guideline Lab, Fact Vault with expiry dates, override-rate telemetry
End audience	Accurate, readable communication	Misled by wrong claims; excluded by low contrast or missing alt text	Indirect: verified claims and WCAG-checked graphics

2.3 Needs that drive the design

- Every flag says why, and which rule or fact it comes from.
- Suggestions keep the creator's purpose and do not rewrite the message.
- The system says when it is unsure instead of asserting a violation.
- Humans decide. Nothing is published or overridden automatically.
- Every decision has an audit trail: who, what, when, and which rule version.
- Confidential brand material is protected, and AI use is disclosed.

3. Research, evidence and what was not done

3.1 Research status

We state this plainly because it affects how the evidence should be read.

Layer	Status for Round 1
Secondary research (reports, papers, regulation, incident coverage)	Complete. Sources in Appendix B; verification status in Appendix A.
Published interviews and talks (desk review)	Complete for the items in section 3.5. These were conducted and published by other people. We did not conduct them.
Competitive review of public product pages	Complete. Public pages only, not hands-on testing (section 3.6).
Baseline experiment on LLM review behaviour	Run by the author: 54 trials on 18 pieces (section 3.2). Small sample, one chatbot, one fictional style kit. Raw scoring sheet to be attached to the repository in Round 2.
Observation log of public practitioner discussions	Six cases recorded by the author (section 3.3). Links and dates are to be attached in Round 2, and the sample is too small to show how common any problem is.
Practitioner interviews (our own)	Not conducted for Round 1. We will not simulate them. The protocol is in section 3.7 and results will be reported in Round 2.

3.2 Baseline experiment: how a plain LLM reviewer behaves

We wanted evidence from our own setting before designing around a problem. We wrote a fictional style kit (Northwind Academy) with rules, a fact ledger and expiry tags, then wrote 18 short marketing pieces, some with planted violations and some clean controls. Each piece was reviewed three times in fresh sessions with the same prompt. 54 rows were scored.

Measure	Result
Pieces whose verdict changed between runs	4 of 18 (22.2%): P07, P09, P13, P14
Planted-rule recall, run 1 / 2 / 3	17/17 (100%) / 13/17 (76.5%) / 14/17 (82.4%)
Planted rules missed in all three runs	None
False positives on clean controls	0
Unparseable outputs	0
Findings that cited a rule not in the guide	1 (P05: a fact ID was written into the rule field)

- **Verdicts flipped.** P07 (missing alt text), P09 (date format), P13 (expired fact) and P14 (shaming tone) moved between flag and pass on identical prompts.
- **Expired facts were approved.** On P13, one run accepted a 12-week course length because the string appeared in the fact ledger, and ignored the tag saying it expired on 30 June 2026.
- **Facts and rules got mixed up.** On P05 the model put a fact ID into the rule field. This is why we require schema-checked output and why every citation must resolve to a real ID.

Limits. One chatbot, one fictional guide, 18 pieces written by us, scored by us. This shows the failure modes are real in our setting. It does not give a general error rate. The 22.2% figure is in the same range as the 13.6% reported independently in [R5], which is consistent but is not confirmation.

3.3 Observation log of public discussions

Six cases from public practitioner discussions, recorded under a short codebook (C1 unclear rules, C2 approval delays, C3 destructive rewrites, C4 automated checks, C5 stale facts and drift). Lines are paraphrased here; verbatim lines of 25 words or fewer, with links and dates, go into the Round 2 repository.

ID	Community	Poster role	What was observed	Code	Design response
O1	r/marketing	Agency account lead	Reviewers raise personal taste as if it were brand compliance, causing revision loops	C1	Override-rate telemetry
O2	r/marketing	Social media manager	Feedback is spread across email, Slack and Docs, and the posting slot passes before sign-off	C2	One role-based workflow

ID	Community	Poster role	What was observed	Code	Design response
O3	r/copywriting	Senior B2B writer	Vague tone adjectives mean little to an LLM; standard tone docs get ignored	C1	Structured rules with FORBIDDEN and MUST forms
O4	r/copywriting	Freelance copywriter	AI tone edits rewrite the whole paragraph and strip the author's cadence	C3	Intent Preservation Metric
O5	r/agency	Agency creative lead	Checking assets before stakeholders see them cuts rejections	C4	Deterministic pre-flight scan
O6	r/marketing	Brand manager	A designer changed a graphic after approval and it ran with expired discount codes	C5	Fact expiry and approved-version fingerprint

Six posts are anecdotes. They shaped the design but are not evidence of prevalence.

3.4 Secondary findings that shape the design

- **LLM judgments vary between runs [R5].** Design consequence: run deterministic checks first, use a model only where judgment is needed, require agreement across repeated and rephrased checks before asserting a violation, and let the system abstain.
- **Accessibility failures are common and mostly machine-detectable [R6].** They map directly onto our deterministic checks.
- **Claims carry a duty to substantiate.** The ASCI Code expects objectively checkable claims to be substantiated [R9]. We treat regulatory rules as optional overlay packs, not core behaviour.
- **Privacy law is arriving.** India's DPDP Rules were notified on 13 November 2025. Consent-manager provisions apply from 13 November 2026. Notice, security-safeguard and breach obligations apply from 13 May 2027, with penalties of up to INR 250 crore for failing to keep reasonable security safeguards [R11]. We design for notice, security and breach handling from the start. This is a design intention, not legal advice.

3.5 Published interviews and talks reviewed

These are interviews by other people. We use them to check whether our design matches what practitioners describe, not as evidence of prevalence.

Source	What it says	How VeriLoom responds	Limit
Callaway / Good Good, Adweek column [R13]	The CEO said approval should never have happened; the columnist says no process existed to flag the video	Risk-based routing to an Authorised Reviewer, with an approval record	Paywalled; read in part
Henry Young interview, Serious Insights [R15b]	"AI slop" is usually a quality judgment; people react to perceived laziness	We do not claim to judge taste (section 6.5)	Single speaker; date not shown in our copy
Breaking and Entering, episode 533, PJ Pereira [R16b]	Backlash to fully AI-generated advertising	Context only; no design change	Episode description only; no quotes used

3.6 Competitive landscape

This is a review of public marketing pages, not hands-on testing. A feature missing from a page is not proof that it is missing from the product, so we state our difference as emphasis and will verify it in Round 2.

Tool	What its public page describes	Not described on the pages we reviewed
Adobe GenStudio for Performance Marketing [R18]	Upload guidelines, validate content, compliance score, approval workflows	Testing the guidelines themselves; time-aware fact checking; abstention
Ziflow [R19]	Proofing, review and approval; content-compliance guidance	Automated source-grounded claim checking
Red Marker [R20]	Flags legal, regulatory and brand risks, including unsubstantiated statements	Guideline linting; exact colour and contrast maths; abstention
Contently [R21]	Style guides, blocklisted words, approval workflows	Semantic or claim verification

3.7 Protocol for primary research in Round 2

- **A. Practitioner interviews.** Five to eight people: college club social leads, campus communications staff, freelancers, small-business owners, agency account managers. Ten to fifteen minutes each. Questions: (1) Describe the last time something went out that should not have. (2) Who reviews your content, and how? (3) Where do the brand rules live, and how do you know they are current? (4) What do you do when you disagree with a reviewer? (5) What would make you trust an automated check? We will get consent, anonymise names and ask for no confidential material.
- **B. Observation log, extended.** Grow the six cases to 10 or more, with links and verbatim lines.
- **C. Baseline repeat.** Repeat section 3.2 with a public, licensed style guide in addition to our fictional kit, and run it again against VeriLoom so the comparison is like for like.

We report small samples as small. If we do not complete a part, we will say so.

4. Proposed solution, value proposition and workflow

4.1 Value proposition

VeriLoom is a pre-publication gate. It reviews content against an organisation's own approved guidelines, facts and assets, explains every finding with its source, suggests fixes that keep the creator's purpose, and leaves every decision to people.

- **Creators** get fast, specific, explained feedback before submission, instead of a late and vague rejection.
- **Managers and reviewers** see evidence and only the risky or uncertain items, with a full revision and audit trail.
- **Brand admins** learn which rules are unclear, ignored or contradictory, and can test a rule change before it affects anyone.
- **The organisation** gets a record of who approved what under which rules, and a plain disclosure of where AI and external services were used.

4.2 What makes the approach different

Common approach	VeriLoom approach	Why it matters
Chunk the PDF into 500-token pieces and ask an LLM to score content	Compile guidelines into a Brand-AST: structured rules with a modality (MUST, FORBIDDEN, CONDITIONAL), an immutable ID and a source clause	Chunking separates rules from their exceptions; unclear rules give inconsistent verdicts whatever the model
Match claims to facts by embedding similarity	Atomic claims checked against facts valid on today's date, with a deterministic modality-inflation rule	Similarity scores "saves up to 20%" and "always saves 20%" as near-identical. Our own test showed an expired fact accepted (section 3.2)
Ask an LLM to rewrite the paragraph	Span-constrained minimum edit, scored by IPM, shown as a tracked change	Whole-paragraph rewrites erase the author's voice (O4)
Ask a vision model to estimate hex codes and contrast	CIEDE2000 colour distance and WCAG luminance maths, computed under the OCR text boxes	Exact, reproducible, and unit-testable
A dashboard of violation counts	Reviewer Override Rate per rule, to separate creator mistakes from unclear rules	Brief asks where guidelines are unclear
A confidence badge with nothing behind it	Verdict tiers (Violation, Advisory, Needs human) composed by code from the citation, temporal validity, stability and risk	Run-to-run variation is documented [R5] and was seen in our test

Linting, replay and override-rate analysis borrow established ideas from software engineering (linters, regression tests) and finance (back-testing). Applying them to brand governance is our contribution.

4.3 End-to-end workflow

- **1. Set up (Brand admin).** Import guidelines (PDF or Markdown), approved facts (CSV or JSON with owner, source and valid-until date), terminology and visual kits. The system compiles guidelines into rule objects with an ID, source clause, check type (deterministic or semantic), severity, examples, owner and effective date. A linter flags contradictions, missing examples, uncheckable rules and overlapping scopes. The admin resolves

them and publishes a versioned rule set.

- **2. Submit (Creator).** Upload text, DOCX, PNG/JPG or PDF. State the purpose and must-keep points. Tag the origin (employee, agency or AI-generated) and the channel.
- **3. Protect.** Personal data and confidential terms are masked before any external model call. An air-gapped local mode (Ollama) is available. Submitted content is treated as data and never as instructions.
- **4. Deterministic checks.** Terminology (Aho-Corasick matching), formatting, palette distance, contrast ratio, alt-text presence, reading level and logo aspect ratio. These run locally and are reproducible.
- **5. Grounded semantic checks.** Extract atomic claims, retrieve approved facts valid today, and assess tone against brand voice anchors. Each semantic check runs with rephrased prompts; disagreement lowers the verdict tier.
- **6. Findings.** Each card shows severity, a plain explanation, the cited clause or fact with its version, a verdict tier and a suggested fix.
- **7. Creator response.** Accept, edit or dispute. Intent Lock rejects suggestions that drop a must-keep point. Disputes go to a human.
- **8. Review.** The manager reviews. High-risk categories (health, pricing, legal) and Needs-human items route to an Authorised Reviewer. A reviewer can grant a Justified Rule Waiver, which requires a reason code.
- **9. Approve.** Only a person can set Approved. The system stores a SHA-256 fingerprint of the approved version and the rule-set version used, in a hash-chained audit log.
- **10. Learn.** The insights view shows recurring violations by team and origin and the override rate per rule. When an admin edits a rule, Policy Replay shows which stored pieces would change verdict before the edit goes live.

4.4 Worked example (fictional data)

Field	Value
Submitted copy	"Our platform cuts onboarding time by 50%. Join the webinar on 12 Nov."
Creator's intent	Announce the 12 November webinar; convey an onboarding benefit
Extracted claim	Quantified performance claim: onboarding time reduced by 50%
Fact Vault match	No approved fact supports 50%. Nearest entry F-014 says under two weeks and expired on 30 June 2026.
Cited rule	Messaging Guideline 4.2 [RULE-CLAIM-01]: quantified claims need an approved fact entry
Verdict	Advisory: unsupported claim and stale source. Not asserted as a violation, because no approved fact contradicts it.
Stability	Three rephrased checks agree the claim is unsupported
Route	Authorised Reviewer, because it is an unhedged performance claim
Suggested fix (tracked change)	"Our platform helps teams get started faster. Join the webinar on 12 Nov."
Intent Lock / IPM	Webinar date kept; benefit kept; five words changed; illustrative IPM about 0.91 (to be measured in Round 2)
Receipt	Deterministic checks ran locally. PII masked. Rule citations hashed into the audit ledger.

4.5 The three named improvements and how a judge can test them

Improvement	What it does	Acceptance test
1. Hedge-sensitive, time-aware claim verifier	Splits copy into atomic claims with a modality class (hedged, conditional, absolute). Looks up facts only where current date is on or before valid_until. A hedged fact used as an absolute claim triggers a modality-inflation flag citing the fact ID.	(a) Every Violation-tier finding cites a rule or fact ID that exists in the active set. (b) A claim with no supporting fact returns Advisory or Needs human, never Violation. (c) An expired fact is never treated as support (P13 replayed as a test). (d) "Up to 30%" published as "30%" is flagged. (e) Precision, recall and override rate reported per check type on a labelled set.

Improvement	What it does	Acceptance test
2. Intent-locked surgical rewriting	Proposes minimal edits to the offending span. Candidates are scored with IPM and must pass the rule validator.	(a) A suggestion that drops a must-keep point is rejected and regenerated. (b) Accepted suggestions appear as tracked changes with a short rationale. (c) No change is applied without an explicit click. (d) Share of suggestions meeting the IPM threshold reported honestly.
3. Measured visual and accessibility checks	CIEDE2000 distance to the brand palette; WCAG contrast computed on pixels under OCR text boxes; alt-text presence.	(a) Contrast ratios match a reference calculator on a test set within rounding. (b) Same image gives the same result on every run. (c) A known off-palette colour is flagged with its measured distance.

4.6 The formulas

Intent Preservation Metric. For a sentence S and a candidate repair S', IPM = 0.65 x cosine(Embed(S), Embed(S')) + 0.35 x (1 - WordEditDistance(S,S') / Length(S)). We show a suggestion only if IPM is at least 0.85 and it passes the rule validator. The weights and the 0.85 threshold are our starting choices. We will tune and report them on the Round 2 test set, and we will report suggestions that fail.

Reviewer Override Rate. ROR(rule) = dismissed flags / total flags. A rule with many flags and a low ROR (we start with under 15%) points to a creator training gap. A rule with many flags and a high ROR (we start with over 45%) is an unclear-guideline candidate for admin review. These cut-offs are hypotheses, to be set from real reviewer data.

Modality inflation. A claim is a tuple of subject, predicate, metric and modality. If the matching fact is hedged and the draft is absolute or superlative, a deterministic rule raises the flag. No model decides this step.

4.7 Supporting capabilities

- Fact Vault with owner, source and valid-until date.
- Provenance-aware routing: content from agencies or AI tools gets stricter claim checks and mandatory human sign-off.
- Risk-ranked review queue.
- AI disclosure receipt on every review.
- Tamper-evident audit log: append-only events chained by hash.

5. Features, technology stack, integrations and data sources

5.1 Feature map and cut line

Feature	Phase	Note
Guideline import and rule compilation (human-confirmed)	MVP	Markdown and text first; PDF text extraction with PyMuPDF
Rule linting	MVP	Part of the Guideline Lab
Deterministic checks for text and images	MVP	Terms, contrast, alt text, palette distance, logo aspect ratio, readability
Fact Vault with expiry	MVP	Validity window and status tags
Claim verifier with verdict tiers	MVP	Improvement 1; single run in MVP
Intent Lock and tracked-change suggestions	MVP	Improvement 2
Colour and contrast inspector	MVP	Improvement 3
Four-role workflow, comments, revision history, audit log	MVP	Human-only approval
AI disclosure receipt; masking before external calls	MVP	
Basic insights and override-rate view	MVP	Full ROR dashboard in Round 3
Policy Replay for deterministic rules	MVP	Semantic-rule replay is Phase 2
Stability gate: repeated and rephrased checks	Phase 2	Round 2: three runs, one rephrasing. Round 3: full battery
Local-model confidential mode (Ollama)	Phase 2	Round 3

Feature	Phase	Note
Regulatory overlay pack (ASCI)	Phase 2	Optional; an aid, not legal advice
Ad-platform parity monitoring, video and audio, third-party integrations	Roadmap	Section 7.6

5.2 Architecture

Layer	Responsibility	Key decision
1. Ingestion	Parse text, DOCX, images and PDF into structured segments	Keep original and normalised versions, both fingerprinted
2. Deterministic engine	Run compiled rules, colour and contrast maths	No model call; reproducible
3. Retrieval	Hybrid keyword and vector search over rules and facts, filtered by validity date	Returns IDs, so findings must cite what was retrieved
4. Semantic checks	Model assesses atomic claims and tone	Content passed as quoted data; output must match a schema
5. Stability gate and verdict composer	Re-run with rephrased prompts, compare, assign tier and risk	The model cannot set the verdict in free text
6. Suggestion engine and Intent Lock	Propose a minimal edit, compute IPM, check must-keep points	Reject any suggestion that violates the lock
7. Workflow and audit	Roles, states, comments, revisions, hash-chained log	Only a human can approve
8. Insights	Aggregate findings, disputes and overrides	Computed from audit data
9. Model gateway	One interface for hosted or local models, with masking and a disclosure receipt	Swappable per tenant

5.3 Technology stack

Exact libraries, models and licences will be finalised and disclosed in Round 2.

Component	Choice	Reason	Fallback
Frontend	Next.js 14 (TypeScript), Tailwind CSS, TipTap editor, HTML5 Canvas	TipTap allows non-destructive inline highlights; Canvas draws image bounding boxes	React with Vite
API	Python 3.11, FastAPI, Pydantic v2, async SQLAlchemy	One runtime for NLP, vision and retrieval; typed schemas enforce structured model output	Node.js
Database	PostgreSQL with pgvector	Audit data and vector search in one store	SQLite for local development
Text engine	FlashText (Aho-Corasick), RapidFuzz, Textstat, PyMuPDF	Fast term matching; PDF text and image extraction	Plain regex
Vision and accessibility	OpenCV, Pillow, scikit-image (CIEDE2000), EasyOCR	Exact colour maths; text boxes for contrast under text	Tesseract OCR
Retrieval	ChromaDB with BM25 hybrid search	Hybrid search avoids number and name mismatches	pgvector only
NLI / semantic	DeBERTa-v3 NLI for entailment where useful; hosted LLM through the gateway for tone, schema-constrained	Model used only where judgment is needed	Local model through Ollama
Model access	Provider-agnostic gateway; hosted default; Ollama local option	Confidentiality choice per tenant	Local only
Auth	JWT with role-based access control	Four roles	Local accounts for demo
Testing	pytest, a golden-set evaluation script, Playwright, axe-core	Reliability and accessibility evidence	Manual keyboard tests
Deployment	Docker on a low-cost host	Simple for one developer	Local Docker only

5.4 Data sources

Data source	Constraint	Use
Fictional Northwind Academy kit (rules, facts, assets)	Written by the author	Demo, tests, baseline experiment
Public sample style guide (candidate: Mailchimp Content Style Guide [R17])	CC BY-NC 4.0 per its site; terms to be re-checked before Round 2	Round 2 baseline repeat, with credit
WCAG contrast and text-alternative criteria; WebAIM failure categories [R6]	Public standards	Deterministic checks
Regulatory overlay: ASCI Code [R9]	Check reuse terms; link and summarise rather than copy	Optional pack
Golden evaluation set	Author-written copy with labelled violations; the incidents in section 1.3 rewritten as test cases	Precision, recall, override rate
Injection test pack	Attack patterns from the OWASP LLM Top 10 categories [R12], written by the author	Red-team tests

No customer data is used to train any model.

5.5 Interface design

- Mobile-first layouts with three breakpoints (phone under 640 px, tablet 640 to 1024 px, desktop above). Findings are single-column cards on a phone; the diff view is stacked before-and-after on a phone and side by side on desktop.
- Our own interface targets WCAG 2.2 AA: keyboard navigation, visible focus, text contrast of at least 4.5:1, severity never shown by colour alone, screen-reader labels, reduced-motion support. We will check with axe-core and manual keyboard tests.

5.6 The most technically challenging part

Producing reliable, cited verdicts from unstable model output. A naive checker can flag a piece on one run and pass it on the next. We saw this directly (4 of 18 pieces, section 3.2), and an independent study reports the same kind of instability [R5]. Our design answers it in four layers:

- **Layer 1, deterministic first.** Terms, colour distance and contrast need no model.
- **Layer 2, schema-constrained output.** Semantic checks must return JSON matching a Pydantic schema, and every citation must resolve to an existing rule or fact ID.
- **Layer 3, code-composed verdicts.** The model never writes the verdict. Code sets the tier from citation existence, fact validity date and claim risk.
- **Layer 4, repeated checks.** High-stakes semantic claims run three times with rephrased prompts. Any disagreement downgrades the finding to Advisory or Needs human.

We will measure precision and reviewer override rate against a labelled set and report misses.

6. Responsible design: privacy, security and human oversight

6.1 Human oversight

Decision point	Who decides	What the system does
Publishing a rule set	Brand admin	Lints the rules and shows replay impact first
Accepting, editing or disputing a suggestion	Creator	Shows the cited source and the Intent Lock check
Escalated, high-risk or Needs-human items	Authorised Reviewer	Routes by risk and uncertainty, with evidence attached
Final approval	Manager or Authorised Reviewer	Only a person can set Approved; the version is fingerprinted
Overriding an AI finding or granting a waiver	Reviewer, with a reason code	Logs the override; overrides feed the ROR view
Privacy and model settings	Brand admin	Applies masking and model choice per tenant

6.2 Threat model

Threat	Example	Mitigation
Prompt injection through submitted or retrieved content (OWASP LLM01 [R12])	A document tells the model to approve it	Content passed as quoted data. Verdict computed by code from schema-validated fields. The model has no tools and no approval permission. An injection test pack is part of the suite.
Sensitive information disclosure (LLM02)	Confidential product names sent to an external model	Masking before any external call; local-model option; disclosure receipt; no training on customer data
Excessive agency (LLM06)	A model takes an action in an external tool	No tool access for models. Only humans change workflow state.
Hallucinated rule or source (LLM09)	The model cites a guideline that does not exist	Every citation must resolve to an existing ID. Uncited findings become Advisory. (Seen in our test, P05.)
Cross-tenant leakage (LLM08)	One tenant's guideline text retrieved for another	Tenant-scoped indexes and query filters, with tests
Alert fatigue	Too many weak flags	Abstention tier, risk-ranked queue, override-rate monitoring
Reviewer over-trust	Approving whatever the AI says	Evidence shown before the verdict; Needs-human tier; nothing applied automatically
Stale or contradictory guidelines	Two rules disagree; a fact has expired	Linting, effective dates, precedence rules, fact expiry
Tampering with history	Editing an approval after the fact	Role-based access; hash-chained, append-only audit log

6.3 Privacy and security

- Tenant-isolated storage, encryption in transit and at rest, data minimisation, retention and deletion controls.
- Breach events are logged so a notification process can be followed.
- The demo uses a fictional tenant and no real brand or personal data.
- Design targets the DPDP Rules 2025 concepts of notice, consent, security safeguards and breach handling from the start [R11]. This is a design intention, not legal advice.

6.4 Disclosure of AI and external services

Every review produces a receipt listing the model used, what data left the system after masking, and which rules fired. The settings page lists every external service in use. Section 8 discloses how AI was used to prepare this document.

6.5 Known limitations

- LLM checks can be wrong or unstable. Tiers, abstention and human review reduce this but do not remove it.
- Compiling prose guidelines into rules needs human confirmation.
- Image checks cover measurable properties (colour, contrast, alt text, aspect ratio). They do not judge taste or meaning.
- The MVP targets English-language content.
- Our baseline test is small and self-scored (section 3.2), and the observation log is six anecdotes.
- The competitor review uses public pages only. Some secondary sources are vendor-funded (Appendix A).
- We have not yet conducted interviews (section 3.1).
- Regulatory packs are aids, not legal advice.
- One developer limits breadth, which is why the cut line in section 5.1 is fixed.

7. Development plan, roles and expected impact

7.1 Team and responsibilities

Member	Role	Responsibilities	Contribution
Nandha K	Sole participant (all roles)	Policy compiler and data model; claim and retrieval engine; vision and contrast pipeline; review workbench frontend; telemetry, privacy and testing; documentation	100% of planning, build, testing and documentation

This is a solo entry. The author holds every role, is accountable for every decision, and can explain or demonstrate any part of the product.

7.2 Phased plan

Phase	Deliverables	Exit check
0. Round 1 (now)	Research, baseline test, evidence and this plan	This document
1. Round 2 sprint 1 (about 25% of build time)	FastAPI service, database, rule schema, guideline importer, upload, finding cards, four roles, audit log	A creator submits text, receives cited deterministic findings, and a manager approves
2. Round 2 sprints 2 and 3 (about 35% and 25%)	Deterministic and semantic checks, Fact Vault and verdict tiers, claim verifier, IPM suggestions, colour and contrast inspector, review UI	Acceptance tests in section 4.5 pass on the sample tenant
3. Round 2 sprint 4 (about 15%)	Multi-format testing with a few real users; primary research from section 3.7	Round 2 research and test report
4. Round 3 (hardening and finish)	Stability gate full battery, ROR dashboard, Policy Replay, Ollama mode, injection tests, accessibility audit, demo video, reliability report	Golden-set report published; limitations documented

Build window. No product code will be written before the official build window opens. Phase 0 is research and planning only.

Cut rule. If time runs short, drop in this order: regulatory overlay, provenance routing, local-model mode, semantic replay. Never drop human approval, citations or the audit log.

7.3 Risk register (author's assessment)

Risk	Likelihood	Impact	Mitigation
Scope too large for one person	High	High	Fixed cut line and drop order
LLM outputs unstable	High	Medium	Deterministic-first design; stability gate; abstention
Interviews not completed before Round 2	Medium	High	Protocol fixed in section 3.7; report whatever is found; state gaps, do not fill them
Evidence challenged by judges	Medium	High	Only traceable sources cited; status in Appendix A; own data labelled as small
Messy real-world files	Medium	Medium	Limit formats; fail visibly with a clear message
Heavy model dependencies (OCR, NLI) slow the demo	Medium	Medium	Deterministic checks run first and alone; models load lazily; fallback to hosted API
Licence problems with sample guides	Medium	Medium	Use credited, licensed guide; check terms before Round 2
Data leak during demo	Low	High	Fictional tenant only

7.4 Expected impact as testable hypotheses

The targets below were set before measurement. We will report misses as well as hits. We do not claim review-time savings or error-elimination figures today.

Hypothesis	Measure	How measured	Target
H1. Deterministic checks are precise on what they cover	Precision and recall per check type	Labelled golden set	Precision of at least 95%
H2. Violation-tier findings are rarely overridden	Reviewer override rate	Reviewer outcomes during testing	Below 20%
H3. Grounded verdict tiers are more stable than free-form chatbot review	Share of pieces with the same tier across three reruns	Same pieces as the section 3.2 baseline (4 of 18 changed)	Better than the baseline
H4. The verifier never accepts an expired fact	Expired-fact cases passed as supported	Test cases built from P13 and similar	Zero

Hypothesis	Measure	How measured	Target
H5. Linting finds real guideline defects, and high-override rules are the unclear ones	Seeded defects found; override rate against defect labels	Sample guide with seeded defects	At least 90% found
H6. Intent Lock preserves must-keep points	Share of accepted suggestions keeping all points	Test set	100%; failures counted as checker bugs
H7. Creators reach a clean submission faster	Time to clean submission with and without the tool	Small usability test	No target; report what we see

7.5 Evidence we will provide in later rounds

A public repository with commit history, the raw baseline scoring sheet, test and evaluation reports, a decision log, a summary of tester feedback, the Round 2 interviews, and a candid limitations section.

7.6 Future scope

- **Post-approval drift detection.** Compare the approved version with the live ad using the stored SHA-256 fingerprint, via ad-platform webhooks. Motivated by platform-side AI edits [R2].
- **Video and audio review.** Local speech-to-text and keyframe sampling to check spoken claims and caption contrast.
- **Guideline pull requests.** When a rule is a high-override candidate, draft a clearer clause for a manager to approve.
- **Regulatory packs.** Sector rules (for example SEBI/RBI disclaimers, health claims, DPDP notices).
- **Integrations.** Figma, Google Drive, Slack, Teams, and a command-line check for docs sites.

8. Tools, AI assistance and integrity statement

AI assistance. Claude (Anthropic) was used for structuring and drafting this document. The author reviewed and edited the content and is responsible for it.

Sources. Sources were retrieved by web search and, where the text was accessible, read in full or in part. Appendix A says which. Where a claim rests on one party's own account, the document says so.

Primary evidence. The baseline experiment (section 3.2) and observation log (section 3.3) are the author's own, small, and will be backed by raw data in the Round 2 repository. No interviews have been conducted or simulated.

Code. No product code has been written for this submission.

Reused material. Any sample style guide will be used under its licence with credit (section 5.4).

Appendix A. Evidence validation ledger

Status levels: Full means the primary text or full report was read. Partial means the main claim was read in an accessible summary or excerpt. Attributed means the claim is one party's account, reported as such. Own means produced by the author.

Ref	Used for	Status	Note
R1	Coca-Cola incident	Full (Fortune article)	Says the ad appears to misattribute. Agency process claim is attributed to the agency.
R2	Meta AI-features terms; Bookshop.org	Partial (several trade reports)	Effective date reported as moved to 20 November 2026. Meta has not directly addressed the liability question.
R3	Equinox	Full (AthleTech News, 29 Jul 2026)	Review question attributed to commenters.
R5	LLM judge instability	Full (abstract and summary)	Single independent researcher; two models, one provider; submitted 23 April 2026.
R6	WebAIM figures	Partial (secondary reports, cross-checked)	Confirm against the WebAIM report before final export.

Ref	Used for	Status	Note
R9	ASCI influencer disclosure figures	Full (press release and summary)	Study of 100 named creators. ASCI Code clauses not quoted.
R11	DPDP Rules timeline	Full (law-firm summaries)	Dates cross-checked. Some sources say 14 May 2027; we use 13 May 2027.
R12	OWASP categories	Full (2025 list)	LLM01, 02, 06, 08, 09 used as named.
R13	Callaway and Good Good	Partial (headline, standfirst, CEO quote)	Paywalled. Used for the approval-failure pattern only.
R15b	Henry Young interview	Partial (page excerpt)	Date not shown in our copy.
R16b	Breaking and Entering episode 533	Partial (episode description)	No quotes used.
R17	Sample style guide	Not re-checked	Licence terms to be checked before Round 2.
R18 to R21	Competitor descriptions	Partial (public pages as seen in search results)	Re-read each page before final export.
Own	Baseline experiment (3.2); observation log (3.3)	Own	Small, self-scored. Raw sheet and links to be attached in Round 2.

Appendix B. References

R1 Beatrice Nolan, "An AI-powered Coca-Cola ad campaign mistakenly invented a book by a famous author", Fortune, 14 May 2025. https://fortune.com/2025/05/14/ai-coca-cola-ad-campaign-invented-fake-book/

R2 Adweek, "Advertisers pull Meta ads that had unwanted AI glow-ups"; PPC Land (22 Oct 2026), https://ppc.land/meta-makes-advertisers-solely-responsible-for-its-ai-ad-edits-from-oct-30/ ; The Keyword, "Meta's Commercial Terms", https://www.thekeyword.co/news/meta-commercial-terms-ai-liability ; Relevant Audience (23 Oct 2026), https://www.relevantaudience.com/meta/meta-commercial-terms-ai-features-advertiser-liability-october-30/

R3 AthleTech News, "Equinox apologizes for AI ad campaign image", 29 July 2026. https://athletechnews.com/equinox-apologizes-for-ai-ad-campaign-image-asian-woman/

R5 Yagubyan, A., "The Coin Flip Judge? Reliability and Bias in LLM-as-a-Judge Evaluation", arXiv:2606.13685. https://arxiv.org/pdf/2606.13685

R6 WebAIM, The WebAIM Million 2026. https://webaim.org/projects/million/

R9 ASCI, Press release: Half-Yearly Complaints Report 2025-26, 11 November 2025. https://www.ascionline.in/wp-content/uploads/2025/11/Press-Release-ASCI-Half-Yearly-Complaints-Report-2025-26.pdf

R11 JSA Prism, "Digital Personal Data Protection Rules, 2025: Operationalising consent, security, and governance obligations", November 2025. https://www.jsalaw.com/?p=55573

R12 OWASP, Top 10 for Large Language Model Applications 2025. https://owasp.org/www-project-top-10-for-large-language-model-applications/

R13 Nataly Kelly, "Callaway and Good Good's Ad Was So Bad, Even AI Audiences Sounded the Alarm", Adweek, 31 August 2026. https://www.adweek.com/brand-marketing/callaway-and-good-goods-ad-was-so-bad-even-ai-audiences-sounded-the-alarm/

R15b Serious Insights, interview with Henry Young. https://www.seriousinsights.net/henry-young-interview

R16b Breaking and Entering, episode 533, interview with PJ Pereira. https://www.castfox.net/podcast/breaking-and-entering-advertising-podcast-121136/episode/533-pj-pereira-on-taking-the-brunt-of-ai-backlash-from-the-industry-181656

R17 Mailchimp Content Style Guide. https://styleguide.mailchimp.com/ (CC BY-NC 4.0; to be checked before Round 2)

R18 Adobe GenStudio for Performance Marketing, brand compliance. https://business.adobe.com/products/genstudio-for-performance-marketing/brand-compliance

R19 Ziflow, content compliance. https://www.ziflow.com/blog/content-compliance

R20 Red Marker, product page. https://www.g2.com/products/red-marker/discuss

R21 Contently, brand compliance. https://contently.com/brand-compliance/